Summary Information	
Project Number	M26
Project Title	ROV Stabilisation
Project Supervisor/Manager	Yilei Zhang
Project Sponsor/Client	Richard Green
Date Prepared	6 March 2026

Project Description

The UC Vision department has developed an ROV capable of following dropper lines in mussel farms with the aim of collecting samples of mussels. The ROV must be stabilised against motion of the mussel farms in which it operates. A stable ROV is an important component for the successful identification and collection of mussels from the dropper lines.

Problem Statement

The ROV currently cannot fully stabilise itself against the motion of ocean currents whilst navigating through the mussel farms. This affects the ability of the ROV to collect mussel samples which is the key goal of the ROV.

Objectives

- To model the movement of the mussel buoys, and thus dropper lines, in the ocean based on specific measured input (e.g. swell height, tide direction).
- Use the above model to generate a stabilisation solution for the ROV so that it can remain fixed with respect to any given point on the dropper line for ease of mussel collection.

Key Assumptions

It is assumed that I will have access to the ROV's previous operational data, the simulation software, and the ROV software.

Scope of Project

In Scope	Out of Scope
• Computer Vision System • Simulation Software • Previous Operational Data	Grabber Arm Thruster System Tether System

Milestones

Description	Date
Researching current ocean model(s) and investigating applicability and/or how they could be manipulated to suit needs.	10/5/26
Utilise research to develop a model of the ocean that can be used for simulation.	20/6/26
Assess model and refine as needed until suitable accuracy is achieved.	1/7/26
Implement software into ROV using the refined model to stabilise while in operation.	10/8/26
Test the software in both simulation and real world to identify points for improvement and implement accordingly.	15/9/26
Present final solution.	12/10/26

Deliverables

Deliverable	Acceptance Criteria	Date
Ocean Model	Must be reasonably accurate to real world and can be tuned based on real world measurements.	1/10/26
ROV Software Changes	Must make measurable improvements to the ROV's current stabilisation capabilities.	1/10/26
Final Solution Presentation	Must outline processes followed and outcomes of this subproject.	1/10/26

Benefits

The benefit of this project is to improve the performance of the ROV's motion and stabilisation to provide the rest of the ROV's systems a better base of operations to identify and collect mussels for sampling.

Project Structure / Key Resources

Key Resources include:

- Simulation Software
- ROV Operational Data
- ROV Software

Interdependencies
This Sub-Project is interdependent with the mussel identification/selection sub project as both relate to the computer vision system, and the efficacy of this project will impact the performance of the mussel ID system.

Initial Risks to project success	
Key Risks	**Mitigation Plan**

Funding Requirements	
At this stage, no funding requirements have been identified. If any are identified throughout the course of this project, they will be raised with the client (Richard Green).	$

Key Stakeholders & Customers

- UC Machine Vision Department
- Richard Green - richard.green@canterbury.ac.nz
- Yilei Zhang - yilei.zhang@canterbury.ac.nz
- Dirk Pons – dirk.pons@ canterbury.ac.nz

Approval by Sponsor/Client	
Approval	Approved / Not Approved
Project Sponsor	Date / Comments
Comments	